

Mehrdad Kashefi

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Education	Western University , London, ON, Canada <i>Neuroscience Ph.D. Candidate</i> Preliminary Thesis Title: “Neural Basis of Sequential Action” Advisors: Dr. Andrew Pruszynski, Dr. Jörn Diedrichsen Iran University of Science and Technology (IUST) , Tehran, Iran <i>M.S. Electrical Engineering - Bioengineering</i> Lorestan University , Lorestan, Iran <i>B.S. Electrical Engineering - Electronics</i>	2020 - now 2017 - 2020 2013 - 2017
Awards	Neural Control of Movement (NCM) travel award International Ontario Graduate Scholarship (OGS-International) Best Oral Presentation at Neuroscience Research Day, Western University Neuroscience Program Travel Award, Western University Graduated with First Rank GPA, IUST Distinguished Student of EE Department, IUST	2024 2023 2023 2022 2020 2019
Publications		
Preprints		
Journal Articles	[7] Shahbazi M., Ariani G., Kashefi, M. , Pruszynski J.A., Diedrichsen J., “Neural correlates of online action preparation”, <i>Journal of Neuroscience</i> .e1880232024 [6] Kashefi M. , Reschechtko S., Ariani G., Shahbazi M., Alice Tan, Diedrichsen J., Pruszynski J.A. “Future movement plans interact in sequential arm movements”, <i>eLife</i> .94485.1,2024 [5] Chung B., Zia M., Thomas K., Michaels J., Jacob A., Pack A., Williams M., Nagapudi K., Teng L., Arrambide E., Ouellette L., Oey N., Gibbs R., Anschutz P., Lu J., Wu Y., Kashefi M. , Oya T, Kersten R, Mosberger A., O’Connell S, Wang R., Marques H, Rita P. Mendes A., Lenschow C., Kondakath G., Kim J., Olson W.,Quinn K., Perkins P., Gatto G., Thanawalla A., Coltman S., Kim T., Smith T., Binder-Markey B., Zaback M., Thompson C., Giszter S., Person A., Goulding M., Azim E., Thakor N., O’Connor D., Trimmer B., Lima S., Carey M., Pandarinath C., Costa R., Pruszynski A., Bakir M., Sober S. “Myomatrix arrays for high-definition muscle recording”, <i>eLife</i> .88551.1, 2023 [4] Codol O., Michaels J. A, Kashefi M. , Pruszynski J.A., Gribble P. “MotorNet: a Python toolbox for controlling differentiable biomechanical effectors with artificial neural networks”, <i>eLife</i> .88591.1, 2023 [3] Codol O., Kashefi M. , Forgaard C.J., Galea J.M., Pruszynski J.A., Gribble P. “Sensorimotor feedback loops are selectively sensitive to reward”, <i>eLife</i> 12:e81325, 2023 [2] Ahmadi A.* , Kashefi M.* , Shahrokhi H, Nazari M.A., “Computer aided diagnosis system using deep convolutional neural networks for ADHD subtypes”, <i>Biomedical Signal Processing and Control</i> , 2021 (* := equal contribution) [1] Kashefi M. , Daliri M.R., “A stack LSTM structure for decoding continuous force from local field potential signal of primary motor cortex (M1)”, <i>BMC bioinformatics</i> ,	

2021

Conference Abstract	Kashefi M. , Diedrichsen J., Pruszynski J.A., “Is there more to sequence learning than better anticipation?”, Neural Control of Movement (NCM), Victoria, Canada, 2023
	Kashefi M. , Ariani G., Diedrichsen J., Pruszynski J.A., “Planning multiple future actions in sequential reaching”, Neural Control of Movement (NCM), Dublin, Ireland, 2022
Talks	Kashefi M. , Diedrichsen J., Pruszynski J.A., “A compositional solution for sequence learning”, Neural Control of Movement (NCM), Dubrovnik, Croatia, 2024
	Kashefi M. , Diedrichsen J., Pruszynski J.A., “Is motor learning merely anticipation?”, Neuroscience Research Day (NRD), London, Canada, 2023
Teaching Workshop	Introduction to Machine Learning, Brainhack Western 2021
TA	Physiology and Pharmacology Laboratory, Western University 2021
	Computational Neuroscience, IUST 2019-2020
	Electronic Circuits, IUST 2018
Outreach	Mentorship Committee, Society of Neuroscience Graduate Students, Western 2022
	Mentoring first-year graduate students in the neuroscience program.
	Thames Valley Science and Engineering Fair, London, ON, CA 2021-2023
	Judge for Grade 4-12 Science Engineering Fair
	Python Camp, IUST 2019
	Introductory python course for middle school students
Referees	Executive Committee Member, ICBME , IUST 2019
	26th International Iranian Conference on Biomedical Engineering (ICBME)
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